Department of Mechanical Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. Mechanical Engineering
Course Code & Title: ME8792 Power Plant Engineering
Name of the Faculty member(s): Dr.V.Sivakumar, ACSP/ Mechanical
Mr.M.Ashok Kumar, AP (SG)/Mechanical

Date of Practice: 23.09.2022

Innovative Practice Description

Unit / Topic: Unit – II / Diesel, Gas Turbine and Combined Cycle Power Plants

Course Outcome: CO2 – The student will be able to describe the layout, construction and working of the components inside a Diesel, Gas and Combined cycle power plant.

Topic Learning Outcome: TLO 7: Describe the components of Diesel and Gas turbine power plant

Activity Chosen: Theory to Practice

Justification: The standby unit of a Diesel engine power plant is available in RIT power house. The students will be able to understand the working of all the auxiliary systems of a diesel engine power plant while the visit is arranged at the time of the service engineer. So it was arranged on the day when Cummins service engineering came.

Time Allotted for the Activity: 50 min

Details of the Implementation:

The Diesel engine power plant consists 1. Air intake system 2. Fuel system 3. Exhaust system 4. Cooling System 5. Lubrication system 6. Engine starting system 7. Governing System and all these auxiliary systems is there in the Diesel Engine stand by power unit which is available in RIT Power house. We scheduled the visit for the day of September 23, 2022, when the Cummins service engineer arrives, in accordance with the information provided by the Power House In-Charge. The service engineer explains in detail all the auxiliary systems of the diesel power plant. The faculty members and students then had the opportunity to ask the service engineer any questions they had after he thoroughly discussed every auxiliary system of the diesel engine power plant.

No. of Students: 24 + 23

PO / PSO mapping for the activity:

Innovative Practice	PO1	PO2	PO5	PSO3
Level of mapping	3	3	1	3
Justification for correlation	Apply basic engineering concept in coal base power plant – Level 3	Solve the problem the student will apply the mathematics, science and engineering fundamentals - Level 3	Solve the problem in Otto cycle using MATLAB Level 1	Analyze the performance of Diesel and Gas turbine power plant - Level 1

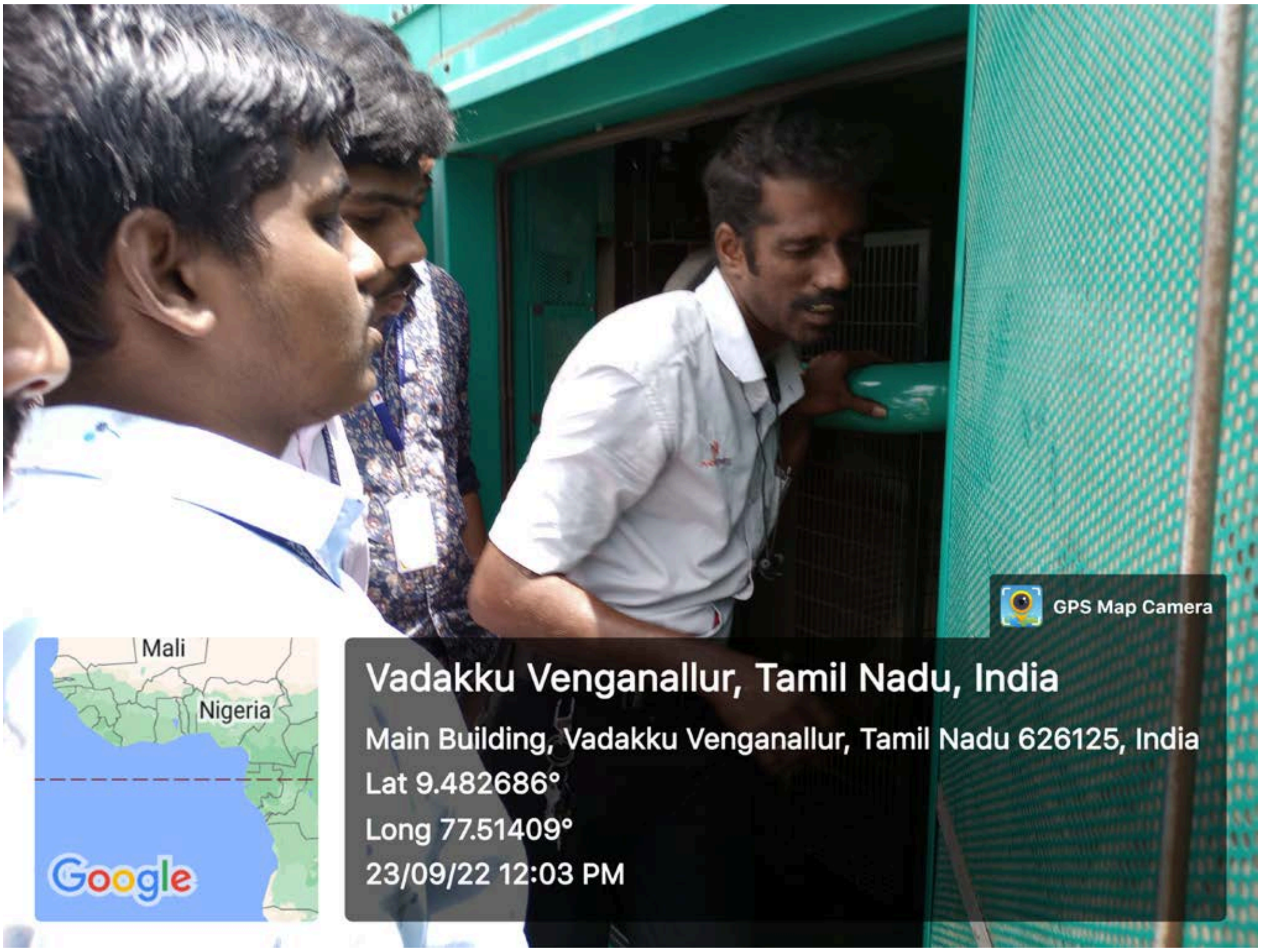
(1 – Low 2 – Moderate 3 – High)

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PSO3
CO2	3	3	1	3

(1 – Low 2 – Moderate 3 – High)

Images / Screenshot of the practice:



Reflective Critique:

- ***Feedback of practice from students and other stakeholders:*** *(samples to be enclosed)*
  Feedback received from students and the sample is enclosed

- ***Benefit of the practice:*** The students can easily understand the working of all auxiliary systems of Diesel Engine Power plant practically.

- ***Whether the practice is adopted in any of the courses early: Yes***
  (If yes provide the details and the modifications you have adopted)

Normally in T2P, the faculty members explain the working and functions of components and the whole system, but this visit was arranged on the date of the service of an engineer from Cummins came because of the fact that the students will be exploring the technology in detail with the specifications of the auxiliary systems.

Challenges faced in implementation: Nil

References:

- Nag. P.K., "Power Plant Engineering", Third Edition, Tata McGraw – Hill Publishing Company Ltd., 2008.

CO2: The student will be able to describe the layout, construction and working of the components inside a Diesel, Gas and Combined cycle power plant.

Signature of Faculty Member **HOD**